

CYBERSECURITY · FOUNDATION

Cybersecurity Foundations

Understand the threat. Build the defense.



Practical security foundations across identity, networks, applications and incident response.



8-12 weeks



4 modules



Professional learning

[Explore the learning experience →](#)

Practical learning. Clear progression. Work you can explain.

YOUR NEXT CAPABILITY

What will you be able to do?

Develop security judgment through threat modeling, IAM, network and application security, cloud concepts, monitoring and incident response exercises.



Understand common attack surfaces



Apply secure development practices



Identify identity and network risks



Practice incident response thinking



Who this is for

- Developers
- QA engineers
- IT professionals
- Security beginners



Your starting point

- Basic IT or software familiarity

Your progression at a glance

FROM CONCEPT TO EVIDENCE



Learn it. Apply it. Explain it.

Move from guided concepts to practical exercises and a coherent piece of work. Use the course resources to revisit difficult ideas and make your reasoning visible.

THE CURRICULUM

A clear route through the subject.

Every module builds towards practical work. The course outline below is aligned with the academy's current program catalog.

1 Security Foundations

Risk, threats and defensive thinking.

- Security principles
- Threat actors and attack surfaces
- Threat modeling
- Security controls

2 Identity and Networks

Secure access and communication.

- Authentication and authorization
- IAM and least privilege
- Network security
- Cloud security basics

3 Application Security

Build and test safer software.

- OWASP risks
- Web and API security
- Secrets and data protection
- Secure development lifecycle

4 Monitoring and Response

Detect, investigate and learn.

- Security logging
- Detection concepts
- Incident response
- Security review capstone

FROM CONCEPT TO EVIDENCE



Practice with purpose

Work within an authorized lab, identify weaknesses, inspect evidence and develop a remediation and incident-response report.

BUILD SOMETHING THAT DEMONSTRATES YOUR LEARNING

Defensive security investigation

Work within an authorized lab, identify weaknesses, inspect evidence and develop a remediation and incident-response report.

**Threat and risk assessment****Lab investigation evidence****Remediation and response
playbook**

Tools and working methods

OWASP

Threat modeling

IAM

Security logging

Tools are part of the learning workflow, not partner endorsements. Examples may evolve while preserving the learning outcomes.



Delivery

Self-paced online · 8–12 weeks

Confirm the current schedule, delivery arrangements and any prerequisites before enrolling.



Completion

A completion certificate is available after the required course work and a passing assessment.

Course completion is not a job, placement or funding guarantee.

Your next steps

- 1 Review the curriculum and your starting point.
- 2 Review current enrollment options and delivery details.
- 3 Start learning, keep your evidence and track your progress.

[Open the course experience →](#)

Customer-preview edition. Interactive links open the local academy prototype. Confirm course availability, schedule and commercial terms with the academy.

INSIDE THE LEARNING EXPERIENCE

Trace an access-control failure in a safe lab

In an isolated training application, changing a project ID in a URL reveals another test user's record. The task is to identify the missing control and verify the repair.

Understand the concept

Authentication establishes identity; authorization determines whether that identity may access a particular object. Enforce the object-level check on every request, including API endpoints that are not directly linked in the UI. Use only the supplied lab accounts and synthetic data. A predictable ID is not the root cause; the absent permission check is.

Worked example

```
GET /projects/42
Session user: learner-A
Record owner: learner-B
Required decision: deny unless an explicit sharing rule grants access
Regression checks: owner allowed; other user denied; no session denied
Log the decision without exposing the private record.
```

Interpret the result

The repaired endpoint evaluates permission before returning the record. Replacing sequential IDs with random IDs may reduce guessing, but it does not replace authorization.

Knowledge check

Does using a UUID instead of an integer remove the need for an ownership check?

Explanation: No. Identifier complexity is not a permission boundary.

YOUR PRACTICAL ASSIGNMENT

Build evidence, not just familiarity.

Trace an access-control failure in a safe lab



Your task

- Create a small threat model covering actors, assets and trust boundaries.
- Write owner/non-owner/no-session regression cases against the local fixture.
- Produce a remediation note with scope, evidence and retest results.



Submission review criteria

- Testing stays within the authorized local lab
- Permission check occurs before private data is returned
- Repair is demonstrated by positive and negative tests

Submit

A working artifact or analysis, a short explanation of your decisions, and evidence that addresses each review criterion.

Reflect

Describe one failed approach, how you diagnosed it, and what you would test next. Identify limitations instead of hiding them.

Your project connection

Work within an authorized lab, identify weaknesses, inspect evidence and develop a remediation and incident-response report.

Use the sample assignment as a stepping stone toward the course project, not as a substitute for the full curriculum.

[Explore the worked lesson](#) →

Proposed teaching sample for curriculum and UX review. Full lesson production, instructor review and delivery scheduling remain separate work.